Removing Indexes On Insert Only Tables

For tables where rows are only inserted and not queried (SELECT), updated, or deleted, these kinds of tables may not have a need for indexes.

For Rideshare, trip_positions is one example of a table where rows are only inserted. To confirm this, run this query from psql:

sql/check_table_dml_behavior.sql

```
-- Check number of inserts, updates, deletes
-- for `trip_positions` table
SELECT
  relname,
  n_tup_ins,
  n_tup_upd,
  n_tup_del
FROM pg_stat_user_tables
WHERE relname = 'trip_positions';
```

If you find Indexes on *Insert Only Tables* and they aren't enforcing a constraint, they may be unneeded and can be removed.

Indexes for PRIMARY KEY and UNIQUE constraints are needed.

For Insert Only tables, since they did not accumulate dead tuples from UPDATE and DELETE operations, in the past PostgreSQL did not VACUUM these tables. Because VACUUM didn't run, table statistics weren't being updated which then could cause incorrect query planning estimates.

That oversight was fixed in PostgreSQL 13 which now performs statistics collection from Autovacuum even for Insert Only Tables.[13] This was mentioned in the *Row Estimates* episode of the Postgres.fm podcast.[14]

You've now seen how Automatic Vacuum jobs should run. What if you want to run manual vacuum jobs or schedule them; how might you do that?

Scheduling Jobs Using pg_cron

pg_cron[15] is a third party extension that has wide support on cloud hosted PostgreSQL databases. You'll work with it in your local PostgreSQL server.

Note that this is the first time you're working with a different database. As a refresher, connect to PostgreSQL using "psql" from your Terminal and run \l to list all the databases. From psql, connect to postgres by running \c postgres.

13. https://www.cybertec-postgresql.com/en/postgresql-autovacuum-insert-only-tables/
14. https://postgres.fm/episodes/row-estimates
15. https://github.com/citusdata/pg_cron